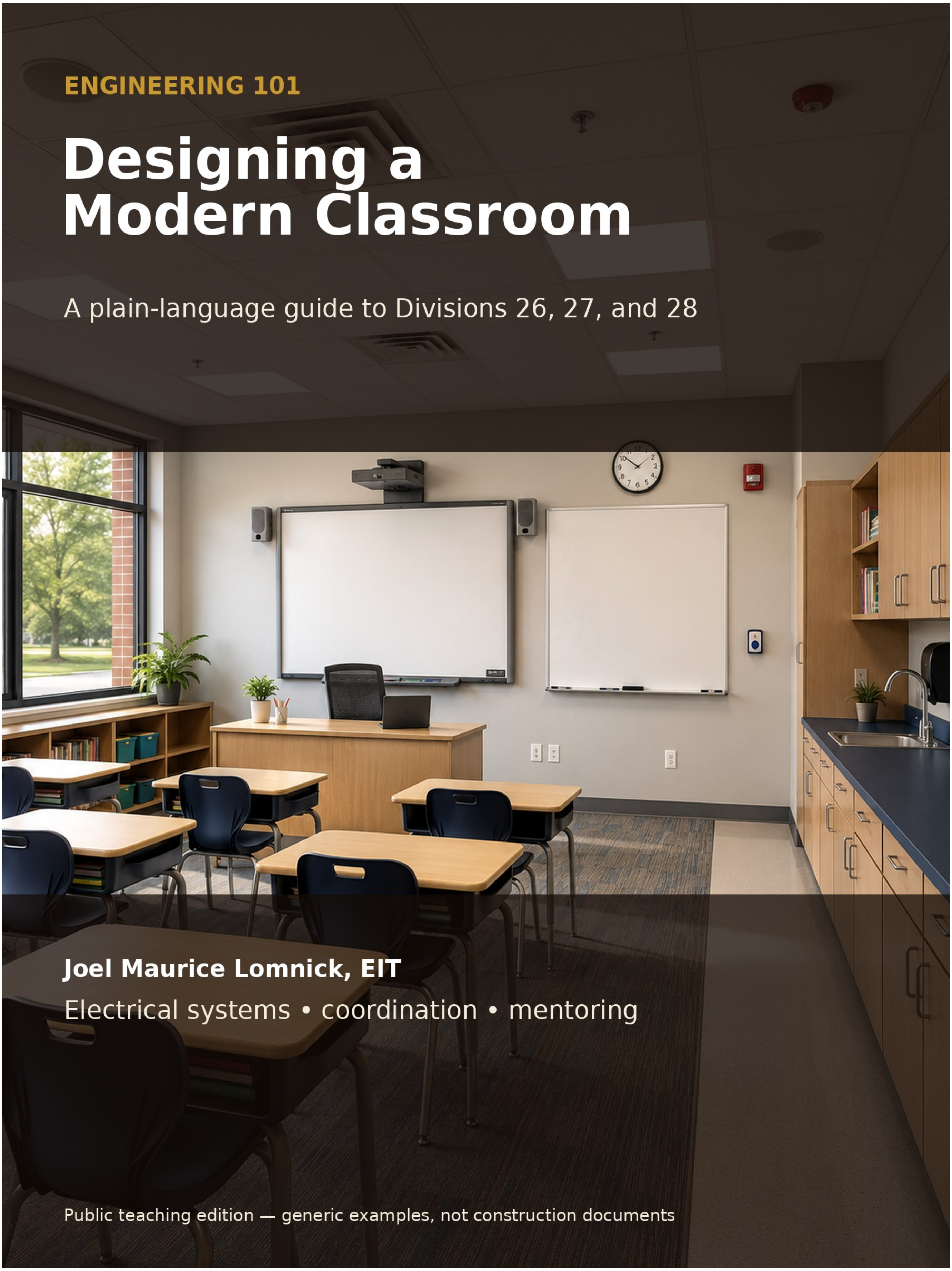


ENGINEERING 101

Designing a Modern Classroom

A plain-language guide to Divisions 26, 27, and 28

Joel Maurice Lomnick, EIT

Electrical systems • coordination • mentoring

Public teaching edition — generic examples, not construction documents

START HERE

A classroom is a small building systems laboratory

Behind a calm school day is a network of power, lighting, communications, life safety, controls, architecture, mechanical equipment, plumbing, and owner technology.

This guide teaches the design logic behind a modern classroom without reproducing project drawings or proprietary details. It is written for students, emerging engineers, owners, teachers, and community members who want to understand what is hidden above the ceiling and behind the walls.

PUBLIC LEARNING EDITION

These examples are intentionally generic. They are not construction documents and are not a substitute for a licensed design professional, the currently adopted codes, owner standards, utility requirements, manufacturer instructions, or the Authority Having Jurisdiction.

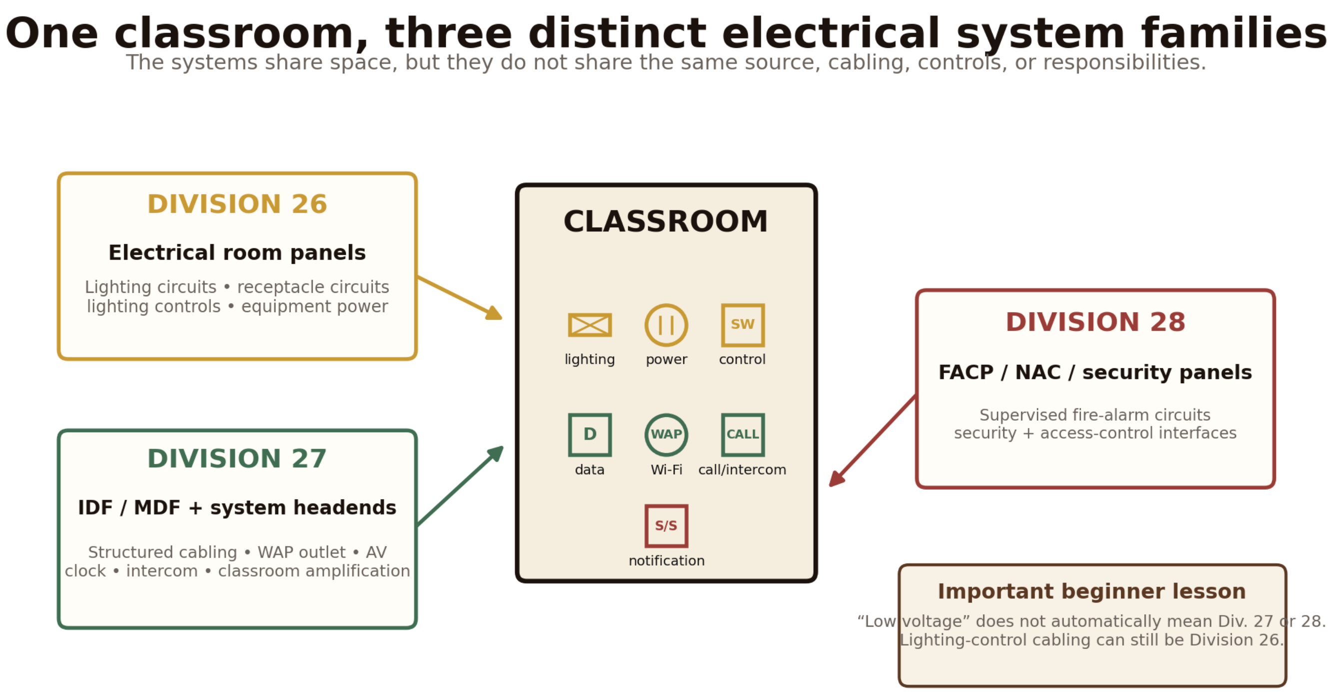
By the end, you should be able to:

- Explain voltage, current, power, energy, circuits, breakers, grounding, and loads in plain language.
- Separate classroom systems into Division 26, Division 27, and Division 28 without mixing their sources or pathways.
- Read basic lighting, power, communications, and life-safety layers as parts of one coordinated room.
- Place common devices on a practice floor plan and explain the design reason behind each location.
- Describe how Revit, AutoCAD, schedules, details, specifications, quality control, testing, and commissioning support the finished design.

THE BIG PICTURE

Start by separating the systems

The same room can contain devices from three electrical divisions. The first teaching skill is learning what serves each device and where its cabling goes.



FIELD REALITY

Project specifications define the exact scope split among the electrical contractor, low-voltage contractor, fire-alarm vendor, security vendor, owner, and equipment suppliers. Never assume responsibility from the symbol alone.

ELECTRICITY BASICS

The few ideas that unlock the rest

You do not need advanced mathematics to begin reading a classroom electrical plan. You do need a disciplined vocabulary.

Voltage (V)

The electrical “push” available to move current through a circuit.

Current (A)

The amount of electrical flow. Conductors and breakers are sized to carry it safely.

Power (W or kW)

How fast electricity is being used at a moment in time.

Energy (kWh)

Power used over time. Utility bills and energy dashboards track this.

Circuit

A complete electrical path from source, through the load, and back to the source.

Breaker

A protective device that opens a circuit when current becomes unsafe.

Grounding and bonding

A safety system that helps exposed conductive parts remain at a safe potential and provides a fault-current path.

Load

Anything that uses electricity: a luminaire, laptop, display, fan, pump, or receptacle-connected device.

BEGINNER HABIT

Never place a device just because there is empty wall or ceiling space. Ask: What serves it? What controls it? What protects it? What other trade occupies the same area? Who must operate and maintain it?

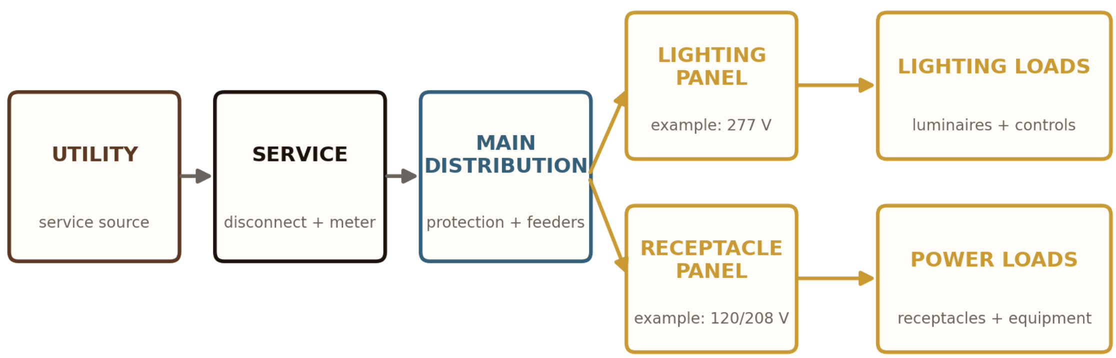
DIVISION 26 — POWER DISTRIBUTION

The engineer starts upstream, not at the receptacle

Utility service, building distribution, transformers, branch panels, fault current, protection, and owner standards shape what can safely reach the classroom.

Division 26: how electrical power reaches the classroom

The classroom is downstream of utility service, building distribution, transformers, and branch-circuit panels.



Example only: actual building voltages and panel arrangements vary. A branch panel normally serves several rooms, not one classroom.

What the classroom commonly receives

- Lighting circuits, often at a voltage selected for building efficiency.
- 120-volt circuits for convenience receptacles and classroom technology.
- Dedicated circuits for equipment that should not share a general-use circuit.
- GFCI protection where the adopted code and equipment conditions require it.
- Clearly identified circuits and panel schedules that maintenance staff can follow later.

PANEL LOCATION

The website activity may let a visitor place a panel to teach circuit origin. In real schools, branch panels are typically in secure, accessible electrical rooms or controlled locations and often serve multiple rooms.

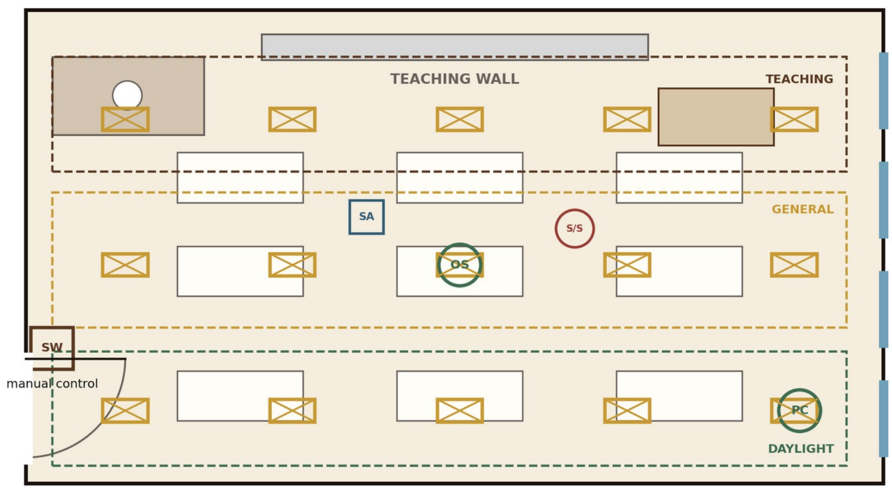
DIVISION 26 — LIGHTING + CONTROLS

Lighting supports several classroom modes

A good layout supports reading, teaching-wall visibility, presentations, daylight, movement, and easy operation — without crowding the ceiling.

Division 26 layer: lighting and controls

Rows of luminaires are coordinated with teaching modes, windows, ceiling devices, and energy-code controls.



Coordination checks

- Board visibility and presentation mode
- Daylight response near windows
- Sensor coverage and commissioning
- Diffusers, sprinklers, speakers, and supports
- Simple controls a substitute teacher can use

Design logic

- Separate the teaching wall, general room, and window/daylight zones when the design benefits from independent response.
- Coordinate manual controls, occupancy or vacancy response, automatic shutoff, and daylight dimming.
- Check sensor coverage around tall furniture, ceiling geometry, doors, and room use.
- Coordinate luminaires with diffusers, sprinklers, speakers, fire-alarm devices, structure, and ceiling supports.
- Commission the controls. A sensor that was installed but never tuned is not a completed design.

ENERGY-CODE LAYER

The exact control sequence depends on the adopted energy code, compliance path, owner standards, and room use. Document the intent clearly enough for programming and commissioning.

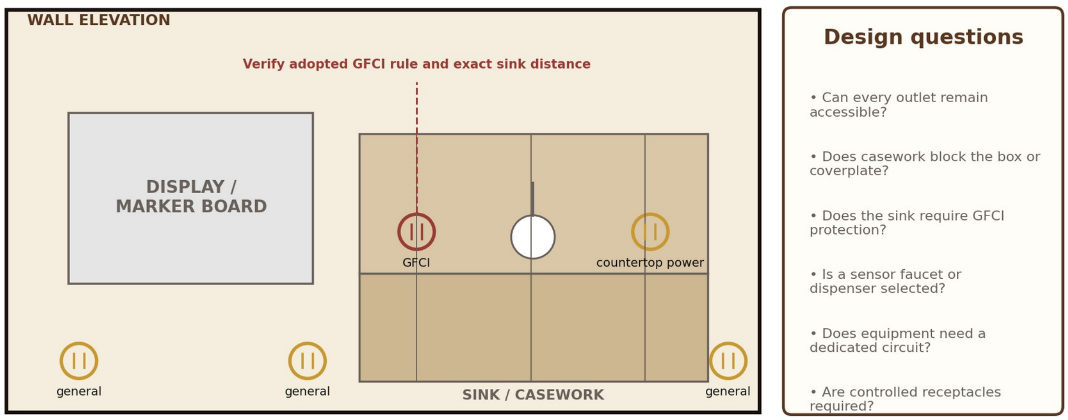
DIVISION 26 — RECEPTACLES + PLUG LOADS

A receptacle plan follows people, furniture, and equipment

The goal is not to fill the walls with outlets. The goal is to reduce extension cords, keep devices accessible, support flexible teaching, and coordinate safely around water and fixed casework.

Division 26 layer: receptacles, casework, and sink coordination

Outlet locations follow furniture and equipment. They are not placed by habit or empty wall space.



Common classroom locations

- General wall receptacles for laptops, chargers, cleaning equipment, and temporary devices.
- Teacher-station power coordinated with data and AV connections.
- Display-wall power coordinated with mounting height, backing, raceways, and final equipment.
- Sink and casework power coordinated with GFCI requirements, cabinetry access, and plumbing equipment.
- Dedicated circuits for copiers, refrigerators, specialty teaching equipment, or other manufacturer-defined loads.
- Controlled receptacles where the adopted energy code and project criteria require automatic shutoff.

OWNER-STANDARD CHECK

Tamper-resistant devices, color coding, mounting heights, and controlled-receptacle arrangements can be driven by owner standards beyond the minimum code. Verify before documenting the final layout.

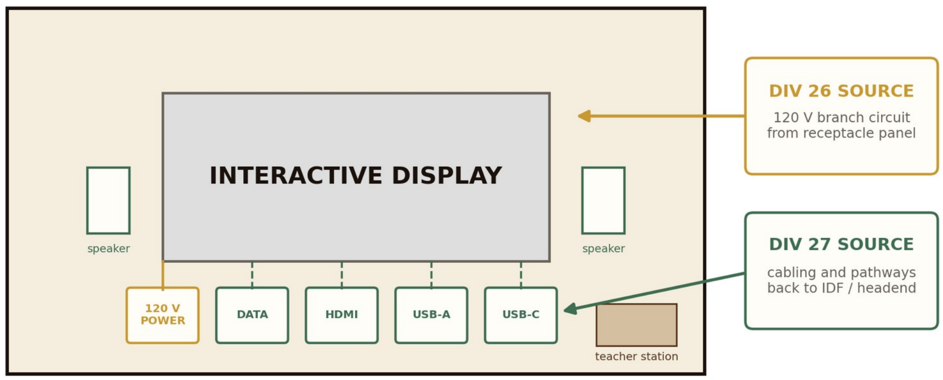
POWER + TECHNOLOGY COORDINATION

The teaching wall is a coordination hotspot

The display may look like one piece of equipment, but its power, data, AV, audio, mounting, and teacher-interface requirements come from several systems and stakeholders.

Teaching wall: power and communications meet, but remain separate

The display needs 120-volt power while its data, USB, HDMI, audio, and network pathways belong to the technology system.



Before rough-in, verify:

- Final display size, mounting height, wall construction, backing, and accessibility.
- Exact quantity and location of 120-volt receptacles.
- Teacher plate or interface ports such as HDMI, USB-A, USB-C, and network data.
- Owner-furnished equipment, technology standards, and who provides patch cords or adapters.
- Conduit routes, bend limits, cable separation, accessible ceiling pathways, and spare capacity.
- Whether audio, classroom amplification, clock, intercom, or call-in systems share the wall.

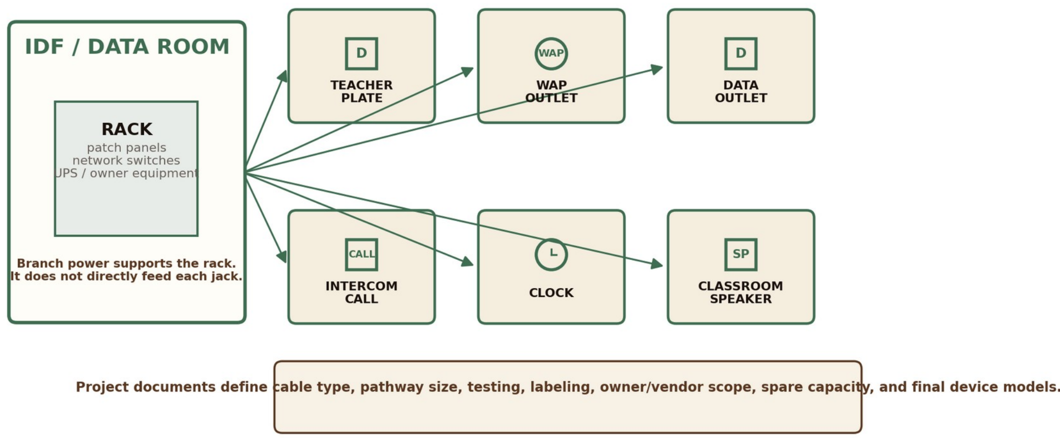
DIVISION 27 — COMMUNICATIONS

The classroom is also a technology space

Structured cabling normally uses a star topology: classroom outlets home-run to an IDF, which connects to the larger school network or system headend.

Division 27: communications home-run to the IDF or system headend

Data and technology devices do not receive branch circuits from the classroom panel. They connect through structured cabling or dedicated system wiring.



Typical Division 27 classroom elements

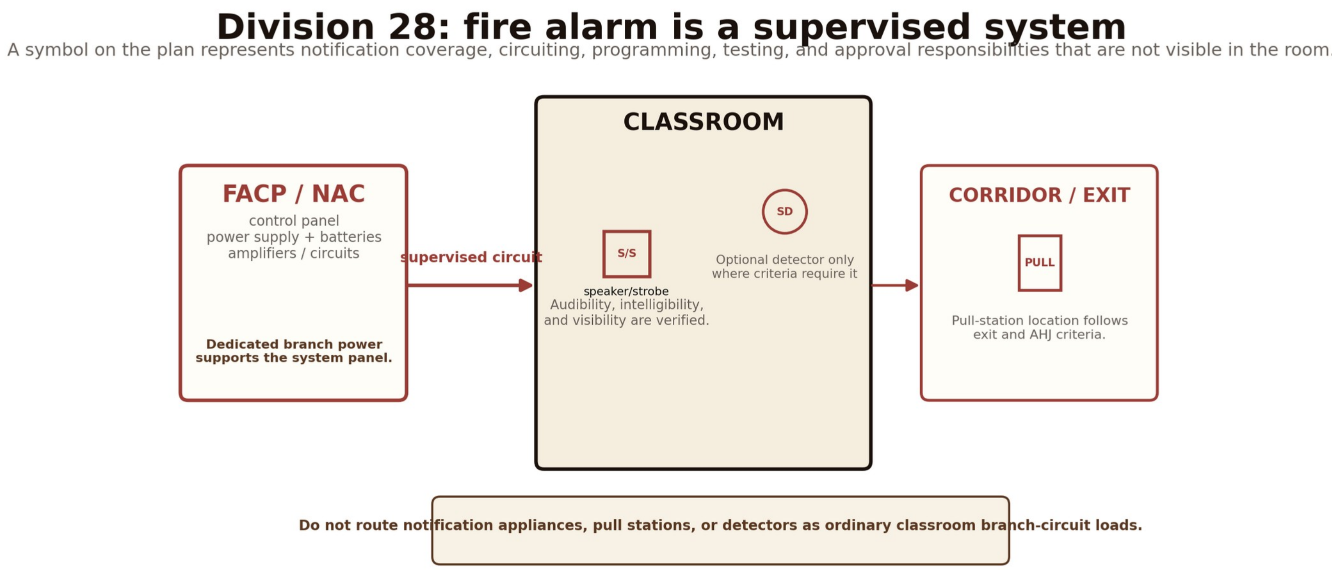
- Data outlets for the teacher station, interactive display, owner equipment, and other defined uses.
- A ceiling or high-wall outlet for the owner’s wireless access point, commonly served by structured cabling and PoE.
- A teacher plate or AV interface with project-specific USB, HDMI, audio, and data ports.
- Classroom amplification, intercom, paging, clock, and staff call-in devices where included by the owner.
- Conduits, sleeves, cable supports, labeling, testing, grounding, and spare pathways back to the IDF or MDF.

SEPARATION MATTERS

The rack may receive 120-volt power or UPS support from Division 26, but the classroom data jack is not a branch-circuit load. The jack connects to the communications network.

Safety devices are designed as a system, not as decorations

A speaker/strobe symbol is only the visible end of a supervised system that includes panels, power supplies, batteries, circuits, programming, testing, and approval.



A classroom-focused life-safety check

- Can occupants see and hear the required notification, and is speech intelligible where voice evacuation is used?
- Is the device location coordinated with ceiling type, wall finish, room geometry, and furniture?
- Are pathways, junction boxes, power supplies, and test points accessible for service?
- Do fire-alarm interfaces correctly release doors, stop equipment, or communicate with other systems where required?
- Has the design separated what Division 26 powers from what the fire-alarm or security system supervises?
- Do the adopted fire code, NFPA 72, owner standards, and AHJ criteria support the selected device type and location?

IMPORTANT RESTRAINT

Do not place a smoke detector in every classroom simply because a symbol is available. Detection requirements depend on the adopted code, system design, room use, and owner criteria.

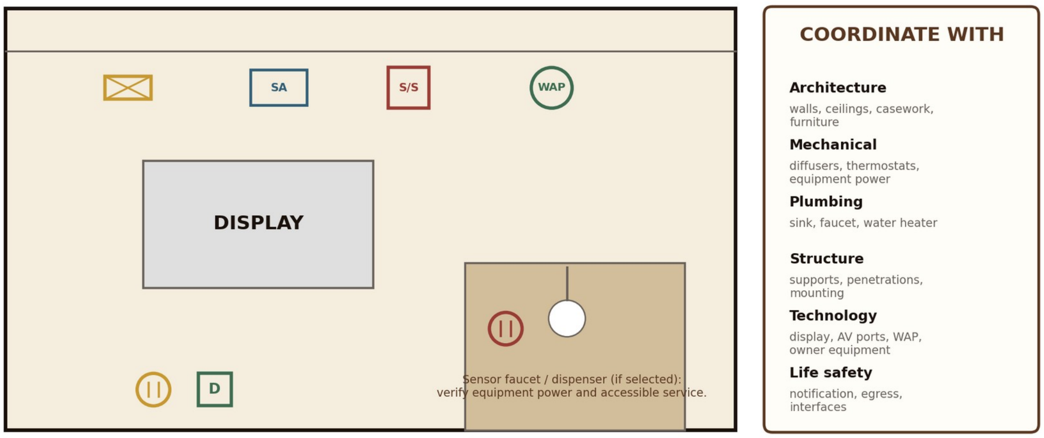
INTERDISCIPLINARY COORDINATION

The electrical plan sits inside everybody else’s plan

The electrical engineer must understand where other systems live and how their equipment changes power, pathways, clearances, controls, and maintenance access.

Coordination: the electrical plan sits inside everyone else’s plan

A correct symbol in the wrong place is still a coordination failure.



SINK EXAMPLE

A classroom may have a manual faucet, or the project may select a hardwired sensor faucet, dispenser, local transformer, or other accessory. The electrical scope follows the actual approved plumbing equipment — not a generic assumption.

Field and construction-manager reality

- Above-ceiling congestion can force late relocations unless the reflected ceiling plan is coordinated early.
- Casework and technology rough-in must be settled before walls close and cabinets are fabricated.
- Core drilling, penetrations, supports, firestopping, and access panels affect several trades and must be assigned clearly.
- Long-lead lighting controls, switchgear, fire-alarm headend equipment, and owner technology can affect schedule.
- Occupied-school work may require night, weekend, holiday, shutdown, temporary-power, or phased-system planning.

ENERGY CODE + COMMISSIONING

The room should use energy intelligently

Efficient fixtures are only one piece. The design also includes automatic shutoff, useful zoning, daylight response, controlled plug loads where required, scheduling, metering, and verification.

System decision	Why it matters	How it is verified
Lighting zones and daylight response	Reduces unnecessary electric lighting and supports teaching modes.	Functional testing, sensor calibration, and sequence review.
Occupancy or vacancy controls	Turns lights off when rooms are unused while preserving intuitive operation.	Walk testing, time-delay settings, and user operation.
Controlled receptacles	Reduces selected plug loads during unoccupied periods where required.	Labeling, control schedule, and functional testing.
Metering and dashboards	Shows owners how building systems use energy over time.	CT ratios, communications, trend data, and commissioning reports.

COMPLIANCE PATH

Confirm whether the project uses a prescriptive or performance-based energy-code path. Controls, lighting power, HVAC efficiency, envelope performance, commissioning, and metering can affect one another.

Commissioning questions

- Do the lighting zones respond as documented, including presentation and daylight conditions?
- Do occupancy or vacancy sensors see the whole room and use appropriate time delays?
- Are controlled receptacles identified and operating on the intended schedule?
- Do meter ratios, network communications, trends, and owner dashboards report believable data?
- Has training been completed so teachers and maintenance staff understand normal operation?

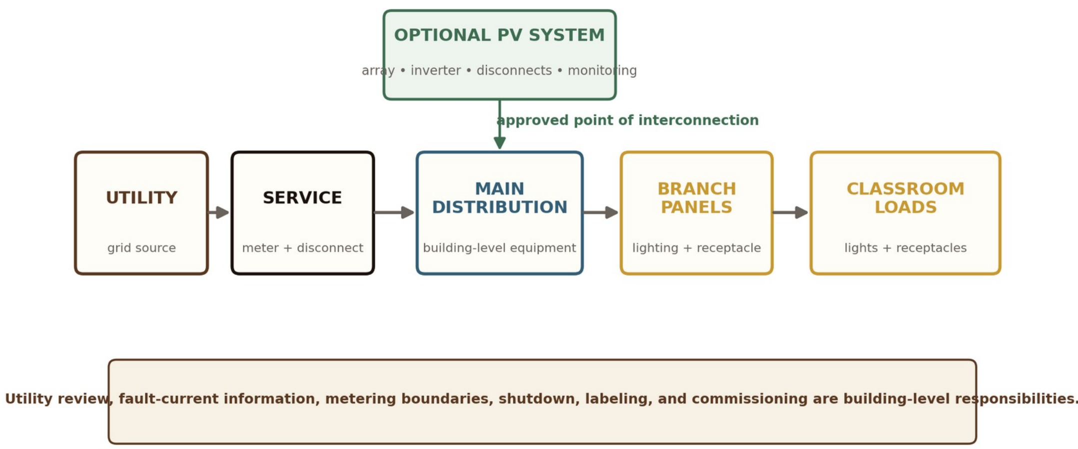
UTILITY + PHOTOVOLTAIC CONTEXT

The classroom is downstream of the utility and the solar array

Utility and photovoltaic coordination happen at the building level. The classroom benefits from the system, but it is not wired directly to the rooftop array.

Building context: utility service and optional photovoltaic generation

The classroom does not connect directly to the solar array. Both are coordinated through the building distribution system.



What a beginner should remember

- Utility service voltage, transformer arrangements, metering, service equipment, fault-current information, and ownership boundaries are coordinated early.
- The photovoltaic point of interconnection must be clear on the one-line diagram and construction documents.
- PV protection, disconnecting means, rapid shutdown, labeling, monitoring, structural support, and utility review are part of the design.
- A classroom plan only shows the final downstream loads. The larger one-line diagram explains how the building safely serves them.

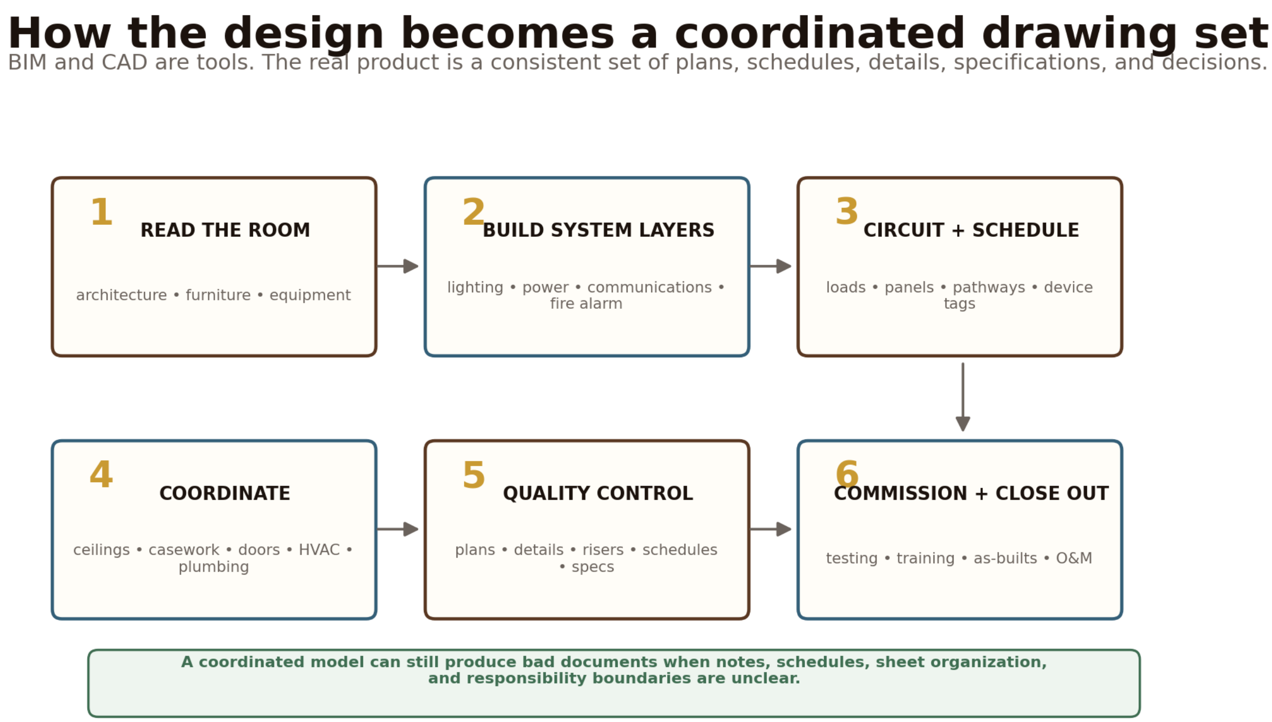
PROCUREMENT AND SCHEDULE

Utility approvals, switchgear, transformers, meters, inverters, and controls can be long-lead or approval-sensitive items. Early coordination protects the construction schedule.

BIM + CAD WORKFLOW

How the design becomes a drawing set

Revit and AutoCAD help organize the work, but the engineer still has to coordinate decisions across plans, schedules, risers, details, specifications, and owner standards.



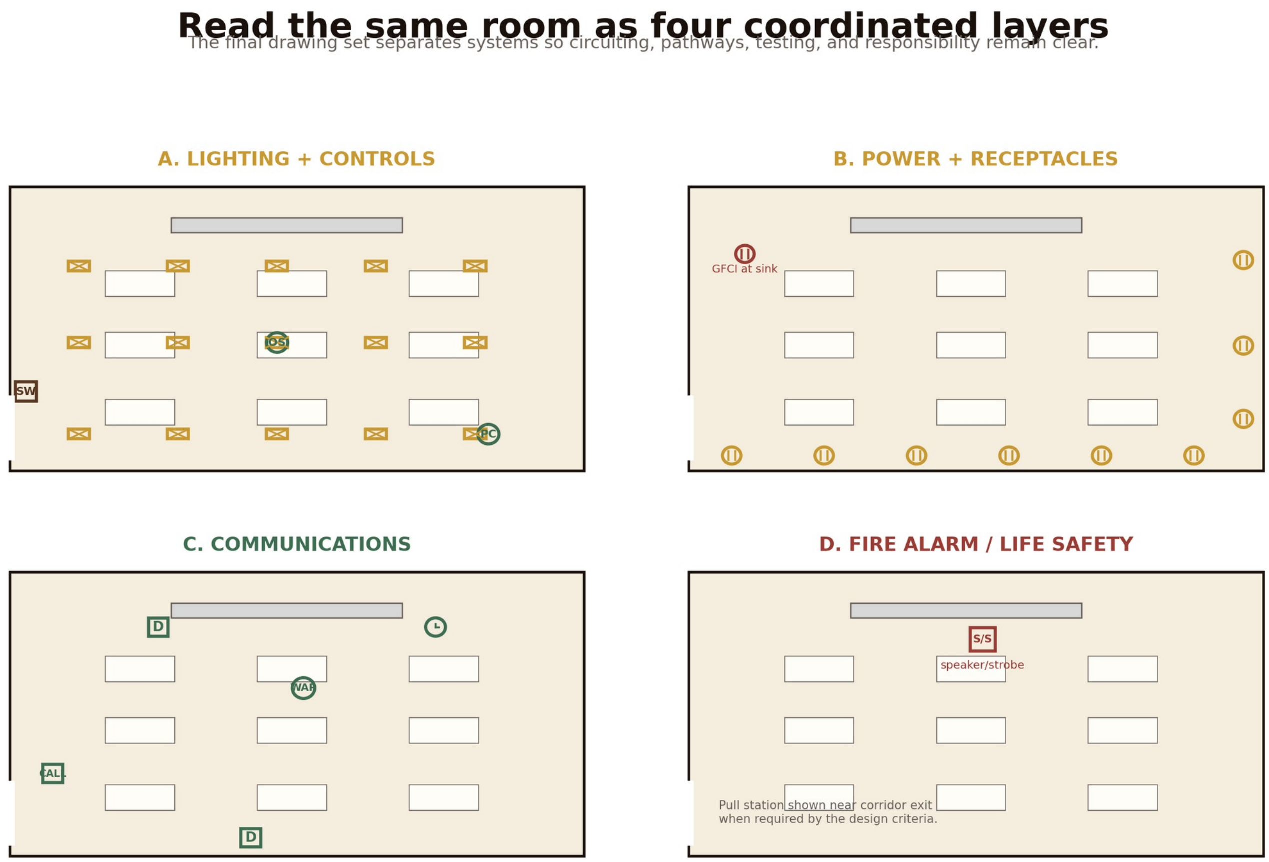
A typical electrical sheet family

- General sheets — legends, notes, schedules, standard details, and system criteria.
- Lighting plans — luminaires, controls, zones, emergency lighting, and reflected-ceiling coordination.
- Power plans — receptacles, equipment connections, panel and circuit information.
- Low-voltage plans — communications, fire alarm, security, clock, intercom, and AV systems as organized by the project.
- Site plans — utilities, ductbanks, exterior lighting, photometrics, and site-system coordination.
- Risers and schedules — one-lines, system diagrams, panel schedules, and equipment relationships.
- Specifications — products, execution, testing, commissioning, training, and closeout requirements that do not fit on the plans.

WORKED EXAMPLE

Read the same classroom as separate coordinated layers

The final construction documents may overlay information differently, but the designer should always be able to explain each system independently.



WHAT THE PICTURE CANNOT SHOW

Circuit numbers, conductor sizes, voltage drop, short-circuit ratings, load calculations, exact code coverage, pathway sizes, equipment schedules, sequences of operation, commissioning, and owner standards belong in the full drawing and specification set.

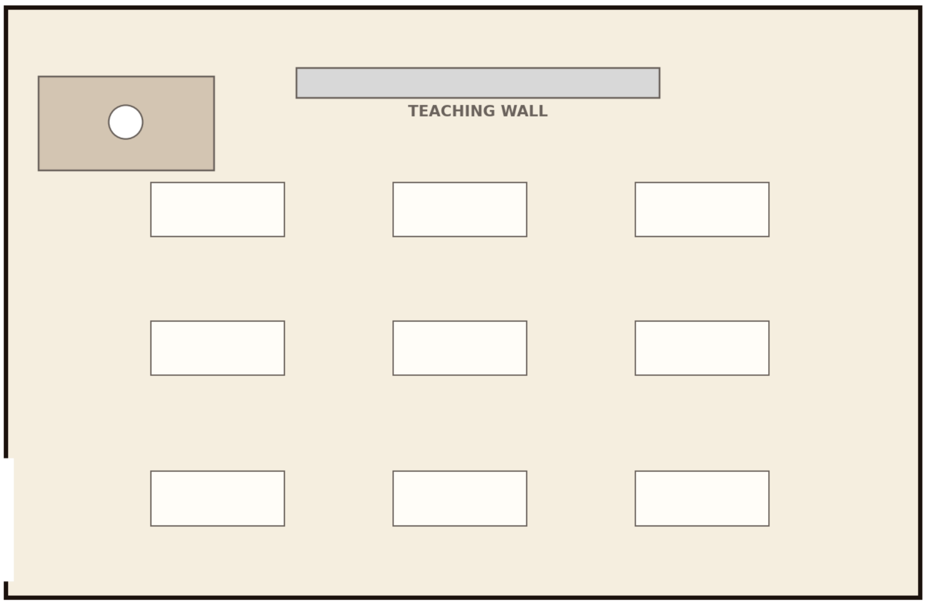
PRACTICE ACTIVITY

Design the room before you reveal the answer

Use the plan as a worksheet. Place the systems in separate layers and write one sentence explaining every important decision.

Practice activity: design the room in separate layers

Place each device on the plan, then identify its Division, source, pathway, control, and coordination partner.



PLACE + EXPLAIN

- 15 luminaires in zones
- manual control by door
- occupancy/vacancy sensor
- daylight sensor at windows
- general receptacles
- GFCI near sink
- teacher power + data + AV
- WAP outlet
- clock and intercom/call
- speaker/strobe
- panel outside classroom
- casework + HVAC coordination

TEACH-BACK PROMPT

For each symbol, say: “This belongs to Division __. It is served from __. It uses __ pathway. It coordinates with __. It is tested or commissioned by __.”

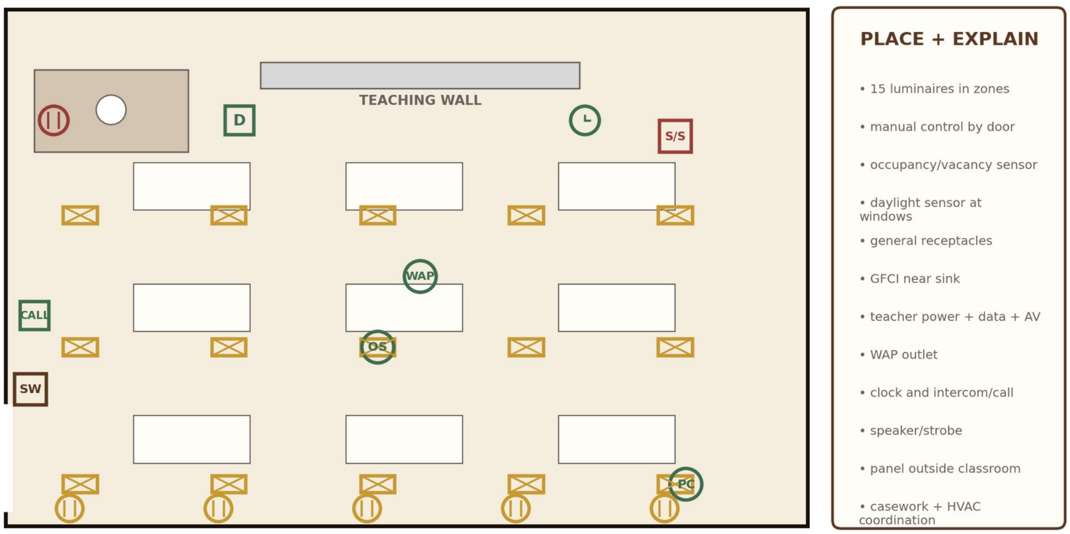
SUGGESTED ANSWER KEY

A good answer explains the logic

There is more than one acceptable layout. The quality of the answer comes from the design reasoning, not from matching every symbol exactly.

Suggested answer key: explain the logic, not just the symbols

A valid answer may differ. The goal is to justify placement, source, controls, pathways, and coordination.



A complete answer should also state assumptions about ceiling type, classroom technology standards, furniture, sink equipment, adopted codes, owner standards, panel locations, fire-alarm system type, and whether the school uses PoE clocks, wireless clocks, or another selected system.

DESIGN SEQUENCE

A ten-step classroom electrical checklist

This sequence keeps the design teachable, coordinated, and buildable.

1. Understand the room

Who uses it? What is taught? What equipment is fixed, movable, or owner-furnished?

2. Read the architecture

Confirm walls, doors, ceiling, casework, furniture, display wall, windows, and accessibility.

3. Build the lighting concept

Choose fixture type, layout, zones, presentation mode, board lighting, and emergency needs.

4. Build the control concept

Define manual control, occupancy/vacancy response, daylight response, and commissioning intent.

5. Place receptacles

Coordinate general, teacher, display, sink, dedicated, controlled, and future-use needs.

6. Place communications

Coordinate data, WAP, teacher plate, amplification, clock, intercom, and call-in devices.

7. Place life-safety devices

Coordinate notification, detection, security, and interfaces with adopted code, owner criteria, and AHJ.

8. Coordinate other trades

Check mechanical, plumbing, doors, ceilings, structure, furniture, technology, and owner equipment.

9. Circuit and schedule

Assign panels and circuits, calculate loads, verify protection, and create accurate schedules.

10. Quality-control the package

Check plans, details, risers, schedules, specifications, responsibilities, and commissioning requirements.

STANDARDS CONFLICT CHECK

Before finalizing: verify the adopted code edition, AHJ policies, owner standards, prototype standards, commissioning impacts, energy-code impacts, and schedule or procurement risks.

QUICK REFERENCE

Classroom device glossary

Panelboard
A distribution cabinet containing buses and circuit breakers for branch circuits.

Branch circuit
Conductors from the final overcurrent device to the outlets or equipment served.

Luminaire
The complete lighting unit: housing, LED source, driver, optics, and mounting parts.

Occupancy sensor
A control input that detects people and supports the programmed lighting response.

Vacancy sensor
Requires manual turn-on, then turns lights off automatically after the room is vacant.

Daylight response
Reduces electric lighting when useful daylight is available.

GFCI
Ground-fault circuit-interrupter protection intended to reduce shock risk.

Controlled receptacle
A receptacle or portion of a receptacle that can be automatically switched off.

IDF
Intermediate Distribution Frame: a telecommunications room serving a building area.

WAP
Wireless Access Point: network equipment that provides Wi-Fi coverage.

Teacher plate
A classroom interface for project-specific AV/data connections such as HDMI, USB, and network.

Speaker/strobe
A fire-alarm notification appliance providing audible voice/tone and visible notification.

NAC
Notification Appliance Circuit or notification power/control equipment, depending context.

One-line diagram
A simplified drawing showing the major electrical distribution path and equipment.

Commissioning
A documented process that verifies systems are installed, programmed, tested, and operating as intended.

AHJ
Authority Having Jurisdiction: the agency or official responsible for interpreting and enforcing applicable requirements.

CLOSING PERSPECTIVE

The best design disappears into a good school day

Students should have enough light to read, enough power to learn, reliable technology, clear announcements, safe emergency communication, and a room that supports different teaching styles.

Teachers should be able to operate the room without fighting it. Maintenance staff should be able to understand, test, and repair it. Contractors should be able to build it without guessing. Owners should know what was installed and how it performs. That quiet reliability is the work.

A note from Joel

Good engineering should be safe enough for the field, clear enough for the owner, and generous enough to help the next engineer learn.

Joel Maurice Lomnick, EIT

Electrical Engineer • Mentor • Community Builder

A note to emerging engineers

Do not measure your value only by how many symbols you can place. Your real value is learning to ask better questions, coordinate across disciplines, communicate clearly, document decisions, and leave behind a system that other people can operate and maintain.

PE Power Exam Tie-In

- Protection and distribution: one-lines, overcurrent protection, transformers, conductors, grounding, and available fault current.
- Lighting and energy: illuminance concepts, lighting power, controls, demand, and energy use.
- Codes and standards: know what must be memorized conceptually versus what must be located in the NEC, NFPA 72, energy code, and owner criteria.

MICRO-PRACTICE QUESTION

A classroom display, WAP, speaker/strobe, and GFCI receptacle are shown on one wall. Identify each device’s Division, source, pathway, required coordination partner, and likely commissioning or testing responsibility.

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